the generation of such a GUI. Details for the above can be found at *https://amarjeetkapoor1.wordpress.com/2016/07/04/user-interface-for-customizing-models/*

A variety of model views. Thorough analysis of a model is very important to arrive at needed precision and accuracy in replicating any 3D structure. To help you achieve this level of perfection, OpenSCAD allows you to view every model in different ways or views.

The initial model view that crops up is CGAL surface view, indicating surface-level representation. It then proceeds to CGAL grid view for complex 3D models. The scaffolds beneath can be seen as a wireframe, creating an effect like the view of Eiffel Tower. If you need a high screen frame rate, this relatively-simple and fast-rendering view should be your choice. You could also have OpenCSG view, or put all of these in one frame for a thrown-together view that gives you as wholesome a look at your design as possible.

Creating animations. An animate option lets you create something similar to GIF files that keep circulating online. Specify the number of steps and frames per second, and you can observe the animation as the program re-evaluates the code as time progresses.

Use with your favourite editor. Be it Emacs, Notepad++, VIM or Visual Studio Coke, minimise OpenSCAD editor, open your own, save after changing the file. OpenSCAD automatically updates its database, letting you smoothly mix and match tools to find the combination that works for you. You can also extend this to work with multiple screens—one for 3D view and another for the code, side by side.

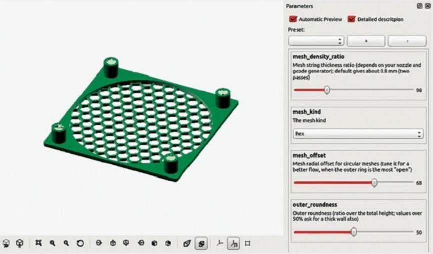

Fig. 2: OpenSCAD customiser

How to work with OpenSCAD

It is clear that this software is different and has some useful tricks up its sleeve. It might seem a little complicated to use at first, but once you know your way around, you will surely enjoy the experience.

Built-in functions. We saw that you do not design, but you code, in OpenSCAD. The script file simply has to make use of functions that are already defined and the graphical image pops up on the screen. Say, you want to draw a cuboid. You would only need to write "cube([2,3,4]);" and then compile the code and go to the graphical view. Preview before you render the image to play around with dimensions or colours. A cuboid of two units length, three units width and four units height appears on the screen, at the origin.

Scrolling around. The UI of OpenSCAD comes segregated into viewing area, console window and text editor. To work around the rendered image, you can use the mouse to rotate, move or zoom in and out of an image. As you scroll, values on the screen/axis change to reflect the action you performed. You can create replicas by simply translating an earlier defined line of code; just take care of positioning, lest you end up overlapping the two. If you are beginning your first program, go into Example under Files and you will find enough codes to get acquainted with the tool.

Model using these techniques. Create complex surfaces or objects using Boolean operators to combine simpler objects, namely, constructive solid geometry (CSG). CSG builds complex forms as intersections, unions and differences of simple primary shapes like boxes, cylinders, cones and ellipsoids. For objects of a fixed cross-sectional profile, you could use extrusion of 2D outlines to your satisfaction.

Specifications at a glance

- Licence: Free software released under GPLv2
- Supported by Linux/UNIX, MS Windows, Mac OS X; it is possible to build OpenSCAD on other systems as long as a C++ compiler and prerequisite software libraries are available
- Built on top of free software libraries like:
  - Qt for user interface
  - CGAL for CSG evaluation
  - OpenCSG and OpenGL for CSG previews, boost, eigen and glew
- Also held at Github
- Latest release: 2015.03
- Website: *http://openscad.org/*
- You can even choose to work with OpenJSCAD, a port of OpenSCAD that runs in a Web browser, if your browser supports WebGL